



Soulmate Sync

AI-Powered Matrimonial Platform

Helping individuals and families discover meaningful life partners through intelligent matchmaking, verified profiles, and cultural compatibility — powered by AI.

Project Presentation

Full-Stack Web Application • Next.js 14 • TypeScript • AI/NLP Engine

July 2026

Project Overview

A modern, secure, and intelligent matrimonial platform for the Indian audience

"Soulmate Sync bridges traditional family values with modern AI technology — making the journey of finding a life partner safe, efficient, and meaningful."

38

APP ROUTES

15+

CORE MODULES

214

USERS SEEDED

201

QA TEST CASES

Target Audience

 **Working Professionals**

Age 24-38, seeking meaningful connections with verified profiles

 **Family Members**

Parents managing profiles for adult children with family account system

 **Urban & Semi-Urban India**

Multilingual support (English, Hindi, Marathi) with 40K+ locations

Architecture & Tech Stack

Enterprise-grade full-stack architecture with AI capabilities

**Frontend**

Framework	Next.js 14 (App Router)
Language	TypeScript 5+
UI Library	React 18
Styling	Tailwind CSS 3
Animations	GSAP 3
Real-time	Socket.io Client
State	React Context API

**Backend**

Runtime	Node.js 20 LTS
Framework	Express.js 4
ORM	Prisma 5
Database	PostgreSQL 15+
Cache	Redis 7+
Auth	JWT (Access + Refresh)
Testing	Jest 30 + Playwright

**Cloud**

AWS ECS Fargate •
Cloudinary CDN •
CloudWatch • Docker

**AI/ML**

Custom NLP Engine •
Intent Classification • RAG
• TF-IDF Scoring

**Integrations**

Razorpay • Firebase OTP
• Twilio SMS • Web Push
API

Core Modules

Complete feature breakdown of the application



Authentication & Onboarding

- ✓ Mobile OTP + Email signup + Google login
- ✓ JWT-based auth with token rotation
- ✓ Multi-step profile creation wizard
- ✓ Photo upload with Cloudinary CDN
- ✓ Biodata PDF extraction with AI
- ✓ 40K+ India locations database
- ✓ Age verification (18+ enforcement)



Profile & Matchmaking

- ✓ Comprehensive profile with 20+ fields
- ✓ AI compatibility scoring (0-100%)
- ✓ Multi-factor matching algorithm
- ✓ Advanced search with 15+ filters
- ✓ Geospatial "Near Me" (Haversine)
- ✓ Profile views & interest tracking
- ✓ Photo gallery with horizontal scroll



Messaging & Communication

- ✓ Real-time chat via Socket.io
- ✓ Text, image & voice messages
- ✓ Typing indicators & read receipts
- ✓ Online/offline status
- ✓ Send/receive interest system
- ✓ Block & report functionality
- ✓ WhatsApp integration



Video & Voice Calling

- ✓ WebRTC-based audio/video calls
- ✓ Call duration timer
- ✓ Mute/unmute & volume controls
- ✓ Number masking for privacy
- ✓ Call history & missed calls
- ✓ Call scheduling interface
- ✓ Voice assistant (speech-to-text)

AI & Premium Features

Intelligent features that set the platform apart



AI Chatbot Engine

- ✓ Multilingual NLP (English, Hindi, Marathi)
- ✓ 15+ intent categories classified
- ✓ RAG with TF-IDF document scoring
- ✓ Entity extraction (names, dates, prefs)
- ✓ Sentiment analysis
- ✓ Context-aware conversation memory
- ✓ Confidence scoring per response



Kundali (Astrology) Matching

- ✓ Astrological compatibility (0-100%)
- ✓ Guna matching (score out of 36)
- ✓ Manglik status detection
- ✓ Nakshatra compatibility assessment
- ✓ Detailed recommendation output
- ✓ Profile-specific results



Premium & Subscription

- ✓ 3-tier plan: Free / Premium / Elite
- ✓ Payment gateway (Razorpay/UPI)
- ✓ Unlimited messaging & video calls
- ✓ See who viewed you
- ✓ Profile boost & priority matching
- ✓ Coupon/discount code system
- ✓ Plan upgrade/downgrade flow



Verification System

- ✓ Mobile & email OTP verification
- ✓ Photo verification (selfie match)
- ✓ ID document upload & verification
- ✓ LinkedIn profile check
- ✓ Verified badge display
- ✓ Admin moderation queue



Relationship Coach

6 interactive coaching modules: First Impressions • Starting Conversations • First Meeting Tips • Family Dynamics • Building Compatibility • Safety & Red Flags — with 5-6 detailed tips per topic and daily affirmations.

Community & Events

Social engagement beyond traditional matchmaking



Community Groups

- ✓ 12+ community groups across 8 categories
- ✓ Regional groups (Tamil, Punjabi, Marathi...)
- ✓ Professional networks (IT, Doctors, MBA)
- ✓ NRI communities
- ✓ Parents-focused groups
- ✓ Support groups (second marriage)
- ✓ Join/leave with member count tracking



Event Matchmaking

- ✓ 7+ featured events
- ✓ Speed dating events
- ✓ Social mixers & coffee meetups
- ✓ Online webinars
- ✓ Festival celebrations
- ✓ Professional meetups
- ✓ Event registration & capacity tracking



Family Account System

- ✓ Role-based permissions (Admin/Manager/Viewer)
- ✓ Add family members (parent, sibling, friend)
- ✓ Profile management delegation
- ✓ Activity tracking per member
- ✓ Safety controls & permission model



Wedding Vendor Marketplace

- ✓ 9 vendor categories
- ✓ Venues, Photography, Catering
- ✓ Decoration, Music, Makeup
- ✓ Jewelry, Invitations, Misc
- ✓ Ratings, reviews & price range
- ✓ Search & category filtering



Host System

Dedicated host portal for matchmakers and community leaders to manage introductions, host meetup events, and facilitate connections between compatible profiles with a separate login and dashboard.

Admin Panel & Security

Enterprise-grade management and protection



Admin Dashboard

- ✓ Analytics: DAU, MAU, revenue, match rate
- ✓ User management (ban/suspend/activate)
- ✓ Moderation queue for reports
- ✓ Verification queue (photo/ID review)
- ✓ Audit logs for admin actions
- ✓ Event & coupon management
- ✓ Feature toggles



Security & Privacy

- ✓ JWT with access + refresh token rotation
- ✓ Rate limiting (5000 req/15min dev, 500 prod)
- ✓ CORS policy enforcement
- ✓ Helmet security headers
- ✓ Input validation (express-validator, Zod)
- ✓ Photo masking (visible only on accept)
- ✓ Number masking for calls
- ✓ GDPR data export support



Mobile & PWA

- ✓ Progressive Web App support
- ✓ Install to home screen
- ✓ Offline capability (Service Worker)
- ✓ Mobile-first responsive design
- ✓ Bottom dock navigation
- ✓ Web push notifications
- ✓ Touch-optimized interactions



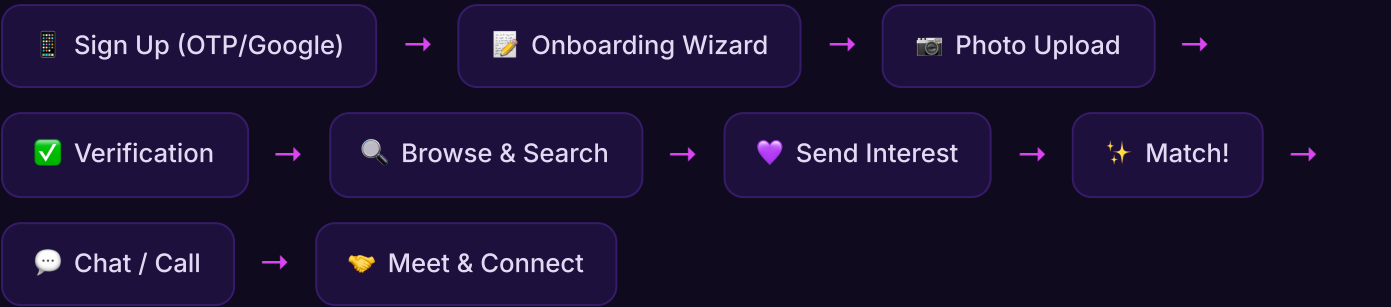
Settings & Preferences

- ✓ Profile visibility controls
- ✓ Privacy settings (who can message/view)
- ✓ Notification preferences
- ✓ Dark/light mode toggle
- ✓ Language switching (EN/HI/MR)
- ✓ Account deactivation option
- ✓ Data download/export

User Flow & Experience

End-to-end journey from registration to match

Registration → Match Flow



Dashboard Experience

- ✓ Profile completion score & prompts
- ✓ Today's Top Picks (AI-curated)
- ✓ Near You (geospatial)
- ✓ Premium & Verified sections
- ✓ Same profession matches
- ✓ Activity stats strip
- ✓ Real-time notification badges

UI/UX Highlights

- ✓ Glass morphism design system
- ✓ Purple/fuchsia gradient theme
- ✓ GSAP slide-in animations
- ✓ Skeleton loading states
- ✓ Toast notifications
- ✓ Dark mode (default) + Light mode
- ✓ Horizontal scroll photo galleries

Key Differentiator: Unlike traditional matrimonial sites, Soulmate Sync combines AI compatibility scoring, real-time communication, cultural matching (Kundali), family involvement features, and a modern mobile-first experience — all in one platform.

Complete Module Architecture

All 15+ modules and their interconnections

 Auth Module /login, /register, /api/auth	 Onboarding /onboarding (5-step wizard)	 Profile /profile, /profile/[id]
 Dashboard /dashboard (home feed)	 Search /search (filters + AI rank)	 Matches /matches (swipe cards)
 Messages /messages (real-time chat)	 Calls /call (WebRTC audio/video)	 AI Chatbot /api/chat (NLP engine)
 Kundali /kundali (astro matching)	 Premium /premium, /checkout	 Family /family (role-based)
 Community /community (groups)	 Events /events (meetups)	 Vendors Wedding marketplace
 Hosts /hosts (matchmaker portal)	 Coach /coach (relationship tips)	 Admin /admin (full CMS)
 Settings /settings (privacy/notifs)	 Notifications /notifications (push)	 Legal /legal (ToS, privacy)

API Architecture

RESTful API design with real-time capabilities



API Endpoints

GROUP	ENDPOINTS
/api/auth/*	Login, Register, Verify, Refresh
/api/profiles/*	CRUD, Search, Upload
/api/activity/*	Matches, Interests, Views
/api/messages/*	Conversations, Send, History
/api/chat	AI Chatbot (NLP)
/api/community/*	Groups, Join, Leave
/api/events/*	List, Register, Details
/api/payment/*	Plans, Checkout, Verify
/api/admin/*	Users, Reports, Analytics
/api/upload	Photo upload (Cloudinary)



Real-time (Socket.io)

- ✓ Chat message delivery
- ✓ Typing indicators
- ✓ Online status updates
- ✓ Call signaling (WebRTC)
- ✓ Push notifications
- ✓ Activity feed updates



Middleware Stack

- ✓ JWT Authentication
- ✓ Rate Limiting (per-IP)
- ✓ CORS (configurable origins)
- ✓ Request logging (Winston)
- ✓ Input sanitization
- ✓ Error handling (structured)



AI/NLP Pipeline



Testing & Quality Assurance

Comprehensive testing strategy across all layers

201

QA TEST CASES

8

E2E SPEC FILES

17

QA MODULES

38

ROUTES PASSING

 **Testing Layers**

✓ **Unit Tests:** Jest 30 for backend services

✓ **Integration:** API endpoint testing

✓ **E2E:** Playwright browser automation

✓ **Manual QA:** 17 detailed test plan documents

✓ **Security:** OWASP Top 10 coverage

 **QA Coverage Areas**

✓ Authentication & session management

✓ Dashboard & profile operations

✓ Search & matchmaking accuracy

✓ Messaging & real-time features

✓ AI chatbot responses

✓ Video calling stability

✓ Payment & premium flows

✓ Navigation & accessibility

 **E2E Test Scenarios (Playwright)**

01

Authentication flows

02

Dashboard operations

03

Search functionality

04

Messaging system

05

Premium features

06

Kundali matching

07

API integration

08

Navigation & UI

Project Summary

A production-ready, AI-powered matrimonial platform

15+

FEATURE MODULES

38

APPLICATION ROUTES

3

LANGUAGES SUPPORTED

Key Highlights:

- ✓ Full-stack application with Next.js 14, TypeScript, Node.js, and PostgreSQL
- ✓ Custom AI/NLP engine with multilingual intent classification and RAG
- ✓ Real-time communication via Socket.io (chat, calls, notifications)
- ✓ Cultural features: Kundali matching, community groups, family accounts
- ✓ Complete premium monetization with payment gateway integration
- ✓ Enterprise security: JWT rotation, rate limiting, input validation, CORS
- ✓ PWA-ready with offline support and web push notifications
- ✓ Comprehensive testing: 201 QA cases, E2E automation, CI/CD pipeline

Thank You

Soulmate Sync — AI-Powered Matrimonial Platform